

Name: _____

Date: _____

Class: _____

MyDesign Portfolio

Element G Initial Template

Prototype Construction of Design #1 (make as many copies, updating the design number, as needed)

Date:

Photos with captions of your full construction process from day 1 of building through to the last day of constructing your final prototype iteration:

Details about all steps of construction (add step numbers as needed), including the tools and manufacturing methods used:

	Details	Tools and Manufacturing Methods used
Step 1	Gathered materials	PVC pipes, balloons, plastic tubing, hot glue gun
Step 2	Built the frame using PVC pipes	Cutting and connecting PVC pipes
Step 3	Attached plastic tubing to the frame for support	Fitting and securing tubing to PVC pipes
Step 4	Added balloons to the holes of the PVC pipe and cut holes	Cutting balloons and attaching them
Step 5	Sealed edges with hot glue	Hot glue gun to reinforce joints and connections
Step 6		
Step 7		

Notes of the progression from your blueprint to your (various) physical prototype(s), including why and how aspects of your design may have changed:

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Final Prototype Construction of Design #2 (change this number to match the design number you're on)

	Details	Tools and Manufacturing Methods used
Step 1	Gather materials	PVC pipes, balloons, plastic tubing, IV needles, hot glue gun, water sources (bottles or containers)
Step 2	Build the frame using PVC pipes	Cutting and connecting PVC pipes to form the structure
Step 3	Attach plastic tubing to the frame for support	Fitting and securing tubing to PVC pipes
Step 4	Attach IV needles to the plastic tubing	Inserting IV needles securely into the tubing and sealing connections with hot glue
Step 5	Set up the water source	Placing bottles or containers filled with water, allowing the plastic tubing to run through them
Step 6	: Create two holes in the PVC pipes for water release	Positioning the holes to direct water flow onto the plants below
Step 7	Seal all edges and connections with hot glue	Using hot glue to reinforce all joints and prevent leaks
Step 8	After sealing, pour water into the system and watch where the water	

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	flows. While observing the water flow, we will make sure there are no leaks, that the water flows in the right direction, and that it doesn't overflow. If there is a leak, we will seal the parts with added glue or find an alternative blocker .	
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Photos with captions of your full construction process from day 1 of building through to the last day of constructing your final prototype iteration:

Details about all steps of construction (add step numbers as needed), including the tools and manufacturing methods used:

Notes of the progression from your blueprint to your (various) physical prototype(s), including why and how aspects of your design may have changed:

List of material details, including item name, description, where it was purchased/obtained (include URL if applicable), cost per unit (if applicable), # of units, total cost (if applicable)

Material name	Description	Where material was purchased	Cost per unit	# units	Total cost per item
Ballon	A balloon is a flexible, inflatable	Dollar stores	\$1	1	1

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	bag, often made of rubber or latex				
Hot glue gun	A Gun that heats up glue and pushes it out and in order to keep something together and then dries up.	Dollar general	\$8	1	\$8
Water bottle	Something that holds water that you could carry with you and drink whenever thirsty	Free	Free	2	Free
Pvc pipe	A pipe that could connect with other pipes and carry liquids throughout them to get it from one place to another	Home depot	\$3.53	6	\$3.53

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Detailed sketches, which might include CAD if you have learned it in your class:

Construction of the Final Prototype (G2)

When you've finished construction of your final prototype, you must be sure that your prototype allows sufficient functionality that you could collect objective, measurable data for all or nearly all of your design requirements. This requires you to look back at your design requirements from Element C. Copy all of your design requirements from Element C into this table and then add how your prototype will allow you to collect objective, measurable data. Note any weaknesses you find or details that will require expert review because they cannot be mathematically modeled or tested,

For example, if you have a design requirement that your design can "transport medicine a distance of 5 meters in 1 minute", is your prototype sufficiently functional such that you can test that? (e.g. can the prototype move the medicine at least a little bit?)

Design Requirement	How will this prototype allow you to collect objective, measurable data?
System must water plants evenly	By observing and timing water dripping from each IV needle and measuring the soil moisture levels under each plant with a moisture meter.
System must be lightweight and portable	By weighing the full system with a scale and confirming it can be easily lifted/moved by one person.
System must minimize water waste	By measuring how much water is poured into the system vs. how much reaches the plants and checking for leaks.

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Testing or Modeling of Attributes (G3, G4) and Addressing Non-Testable Attributes (G4)

Describe all of your prototype's attributes (features, aspects, details, or subsystems). Demonstrate that your prototype is constructed with sufficient functionality that your prototype can test all attributes of your design. This requires you to look back at your initial proposed design details from Element D.

For example, if your initial design details called for wheels and motors, is your prototype sufficiently functional such that you can test out those wheels and motors?

If there are aspects of your design that you believe cannot be physically prototyped or mathematically modeled within the scope and limits of your class, you must later provide evidence that an expert has reviewed your justifications and that you have incorporated their feedback. You do not need to conduct that review in Element G. For now, enter "expert review required" for these attributes in the final column.

Attribute	Description	Describe your prototype's functionality to test this attribute



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Addressing Non-Testable Attributes (G4)

For all attributes that you listed "expert review required" in the table above, provide a justification for why that is true. "Field experts" are professionals who spend at least a portion of their time working in a field directly related to your project. These field experts are not permitted to double count as stakeholders.

Attribute	Justification for why this attribute cannot be modeled or tested with your prototype